Garansh R DSA 8420 Due 3/25

1. Write down the dual problem of the following linear programs. (You can use Table 1 at the end of the homework for help.)

   (a) (10 points)

$$
\begin{array}{llr}
\max & 5x_1 + 6x_2 + 4x_3 & \\
s.t. & -x_1 + 2x_2 & \leq 1 \\
& 3x_1 - x_2 & \geq 2 \\
& 3x_2 + x_3 & = 3 \\
& x_2 \geq 0,\ x_3 \leq 0 &
\end{array}
$$

$\leq 1 \rightarrow$ Dual var $y_1 \geq 0$

$\geq 2 \rightarrow$ Dual var $y_2 \leq 0$

$= 3 \rightarrow$ Dual var y_3

$\Rightarrow \min\ 1y_1 + 2y_2 + 3y_3$

X_1: $-y_1 + 3y_2 \geq 5$

X_2: $2y_1 - y_2 + 3y_3 \geq 6$

X_3: $y_3 = 4$

$\rightarrow y_1 \geq 0,\ y_2 \leq 0,\ y_3$ free

$\rightarrow$ Dual Problem:

$$
\begin{array}{c}
\min\ y_1 + 2y_2 + 3y_3 \\
\text{s.t.} \\
-y_1 + 3y_2 \geq 5 \\
2y_1 - y_2 + 3y_3 \geq 6 \\
y_1 \geq 0,\ y_2 \leq 0,\ y_3 = 4
\end{array}
$$

(b) (10 points)

$$\begin{aligned} \min \quad & x_1 + 2x_2 - 3x_3 \\ s.t. \quad & -3x_1 + x_2 + 2x_3 = 16 \rightarrow y_1 \text{ (free)} \\ & 2x_1 + 4x_2 + 3x_3 \geq 20 \rightarrow y_2 \leq 0 \\ & x_2 + 2x_3 \leq 6 \rightarrow y_3 \geq 0 \\ & x_1 \geq 0, \; x_2 \leq 0 \end{aligned}$$

Dual Vars

Dual Prob:

$\Rightarrow \max \quad 16y_1 + 20y_2 + 6y_3$

s.t.

$-3y_1 + 2y_2 \leq 1$

$y_1 + 4y_2 + y_3 \leq 2$

$2y_1 + 3y_2 + 2y_3 \leq -3$

y_1 (free), $y_2 \leq 0$, $y_3 \geq 0$

2. Consider the following linear program.

$$\begin{aligned} \max \quad & 3x_1 + 10x_2 + 5x_3 + 11x_4 + 6x_5 + 14x_6 \\ s.t. \quad & x_1 + 7x_2 + 3x_3 + 4x_4 + 2x_5 + 5x_6 = 42 \\ & x_1, \ldots, x_6 \geq 0 \end{aligned}$$

(a) (4 points) Write down the dual problem.

(b) (4 points) Solve the dual problem by observation/simple logic.

(c) (4 points) With your solution to (b), what is the optimal value of the primal problem? Why?

Dual Problem: $\min \; 42y$

s.t.

$y \geq 3$

$y \geq \frac{10}{7}$ $(1.42) < 3$ X

$y \geq \frac{5}{3}$ $(1.66) < 3$ X

$y \geq \frac{11}{4}$ $(2.75) < 3$ X

$y \geq 3$ $(3) \geq 3$ ✓

$y \geq \frac{14}{5}$ $(2.8) < 3$ X

$y \geq 0$

optimal $\bar{y} = 3$

$\rightarrow 42(3) \Rightarrow 126$

3. Consider the following primal-dual pair.

$$\begin{array}{llr} \max & x_1 + 2x_2 + 5x_3 + x_4 & \\ s.t. & x_1 + 2x_2 + x_3 - x_4 \le 20 & \\ & -x_1 + x_2 + x_3 + x_4 \le 12 & \text{(P)} \\ & 2x_1 + x_2 + x_3 - x_4 \le 30 & \\ & x_1,\ x_2,\ x_3,\ x_4 \ge 0 & \end{array}$$

$$\begin{array}{llr} \min & 20\pi_1 + 12\pi_2 + 30\pi_3 & \\ s.t. & \pi_1 - \pi_2 + 2\pi_3 \ge 1 & \\ & 2\pi_1 + \pi_2 + \pi_3 \ge 2 & \text{(D)} \\ & \pi_1 + \pi_2 + \pi_3 \ge 5 & \\ & -\pi_1 + \pi_2 - \pi_3 \ge 1 & \\ & \pi_1,\ \pi_2,\ \pi_3 \ge 0 & \end{array}$$

(a) (4 points) Show that $\bar{\mathbf{x}} = (10, 0, 16, 6)$ is feasible to (P). What is the objective value of (P) at $\bar{\mathbf{x}}$?

$\bar{x} \geq 0$ holds

Constraint

P1: $x_1 + 2x_2 + x_3 - x_4 \le 20$

$\rightarrow 10 + 2(0) + 16 - 6 \le 20$

$20 == 20$ holds

P2: $-x_1 + x_2 + x_3 + x_4 \le 12$

$\rightarrow -10 + 0 + 16 + 6 \le 12$

$12 == 12$ holds

P3: $2x_1 + x_2 - x_3 - x_4 \le 30$

$\rightarrow 2(10) - 16 - 6 \le 30$

$-2 \le 30$ holds

Optimal Value (P) of $\bar{x}$: $10 + 2(0) + 5(16) + 6 = 96$

(b) (4 points) Show that $\bar{\pi} = (0, 3, 2)$ is feasible to (D). What is the objective value of (D) at $\bar{\pi}$?

Subbing $\bar{\pi}$ to constraints:

1: $0 - 3 + 4 \geq 1 \rightarrow 1 \geq 1$ ✓

2: $2(0) + 3 + 2 \geq 5 \rightarrow 5 \geq 5$ ✓

3: $0 + 3 \geq 5 \rightarrow 3 \not\geq 5$ ✗

$\rightarrow \bar{\pi}$ not feasable

(c) (4 points) Use the above results to explain why $\bar{\mathbf{x}}$ is optimal to (P) and $\bar{\pi}$ is optimal to (D).

Assuming $\bar{\pi}$ is actually feasible, then ...

Primal Obj. $= 96$

Dual: $20(0) + 12(3) + 30(2)$

$\Rightarrow 96$

Since $96 = 96$, both solutions would be optimal if $\bar{\pi}$ was feasible.

(d) (6 points) Verify that the complementary slackness conditions are satisfied by $(\bar{\mathbf{x}}, \bar{\boldsymbol{\pi}})$.

Primal Constraint Slacks:

P1: $10 + 2(0) + 16 - 6 = 20$

$\Rightarrow 20 = 20 \quad \pi_1 = 0 \rightarrow \pi_1(0) = 0$ holds

P2: $-10 + 0 + 16 + 6 = 12 \quad \pi_2 = 3 \rightarrow 3(0) = 0$ holds

$\Rightarrow 12 = 12$

P3: $2(10) + 0 - 16 - 6 = 30$

$\Rightarrow -2 = 30$

$\rightarrow$ Slack $< 0 \Rightarrow \bar{x}$ not optimal

$\rightarrow$ Complimentary slackness conditions not satisfied by $\bar{x}$ $\Rightarrow$ solution not feasible or not optimal